

# Homework Set 4, CPSC 6430/4430

LastName, FirstName

**Due 04/10/2026, 11:59PM EST**

## ***K*-NN and Decision Boundary**

The table below provides a training dataset containing six observations, three predictors and one qualitative response variable.

| Obs. | $X_1$ | $X_2$ | $X_3$ | Y     |
|------|-------|-------|-------|-------|
| 1    | 0     | 3     | 0     | Red   |
| 2    | 2     | 0     | 0     | Red   |
| 3    | 0     | 1     | 3     | Red   |
| 4    | 0     | 1     | 2     | Green |
| 5    | -1    | 0     | 1     | Green |
| 6    | 1     | 1     | 1     | Red   |

Suppose we wish to use this data set to make a prediction for Y when  $X_1 = X_2 = X_3 = 0$  using  $K$ -nearest neighbors ( $K$ -NN).

1. Compute the Euclidean distance between each observation and the test point  $X_1 = X_2 = X_3 = 0$ .
2. What is our prediction with  $K = 1$  and why?
3. What is our prediction with  $K = 3$  and why? Please determine the probability assigning to each class (Red or Green) by using uniform weight or distance weight (where each neighbor's weight is determined by  $w_i = \frac{\exp\{-d_i^2\}}{\sum_{j=1}^K \exp\{-d_j^2\}}$  and  $d_i$  denotes the Euclidean distance of testing data to  $i$ -th nearest neighbor data) respectively.
4. Now if we only use the first two features  $X_1$  and  $X_2$ , please scatter plot the six observation points and plot the contour for decision boundary by referring to here.

## Objective for $k$ -means

Given the objective of  $k$ -means:

$$\min_{\mu_1, \mu_2, \dots, \mu_k} \sum_{i=1}^k \sum_{x \in \mathcal{C}_i} \|x - \mu_i\|_2^2. \quad (1)$$

Please prove that  $k$ -means algorithm will monotonically make the above objective non-increasing.

If we change the objective to:

$$\min_{\mu_1, \mu_2, \dots, \mu_k} \sum_{i=1}^k \sum_{x \in \mathcal{C}_i} \|x - \mu_i\|_1, \quad (2)$$

please design an algorithm (similar to  $k$ -means) to make the objective non-increasing and prove it.

Check the Notebook for  $k$ -means

Check the Notebook for  $K$ -NN